Kompyuter tarmoqlariga umumiy nazar

Computer Network Overview

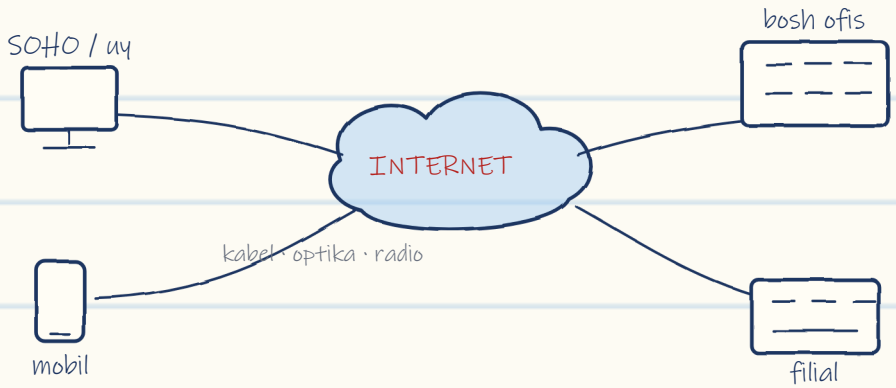
Asosiy atamalar

- Tarmoq — bog'langan, ammo mustaqil kompyuterlar guruhi
- LAN · MAN · WAN — qamrov: bino · shahar ~10 km · davlat
- Topologiya — tugunlarning ulanish shakli (shina, yulduz, halqa)
- Kommutatsiya — kanal band qilinadi yoki paket saqlab uzatiladi
- O'tkazuvchanlik · kechikish — quvur eni (bit/s) va yo'l vaqti

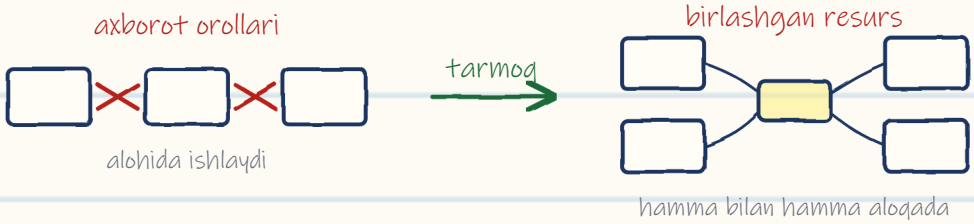
1. Tarmoq nima?

O'zaro bog'langan mustaqil kompyuterlar guruhi (IEEE)

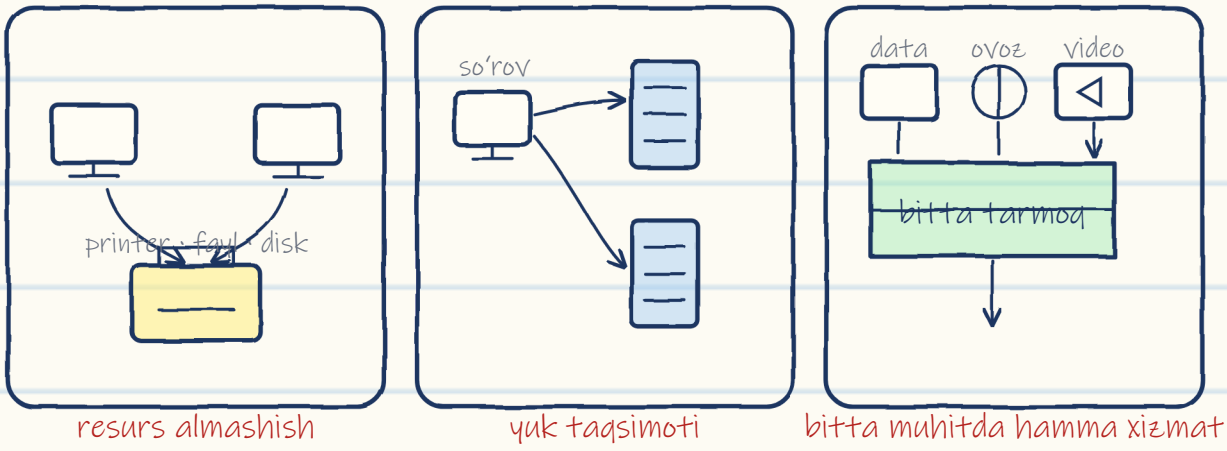
- bog'langan — aloqa muhiti bor, ma'lumot almashadi
- mustaqil — biri ikkinchisini boshqarmaydi



1-rasm. Alohida nuqtalar — bitta tarmoq

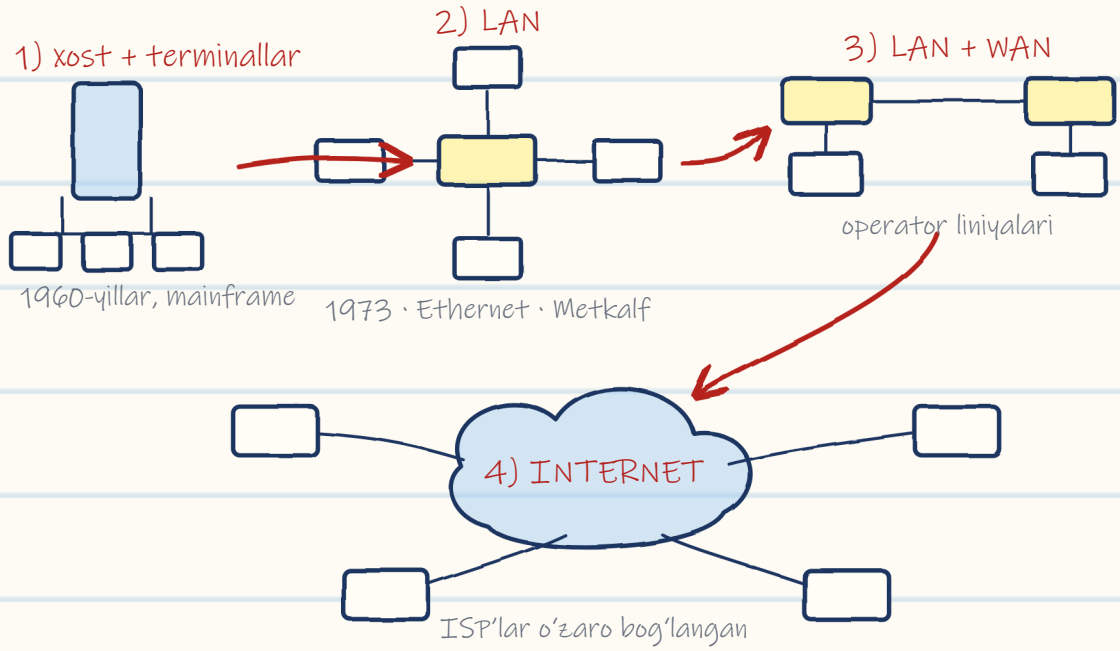


2. Asosiy vazifalari



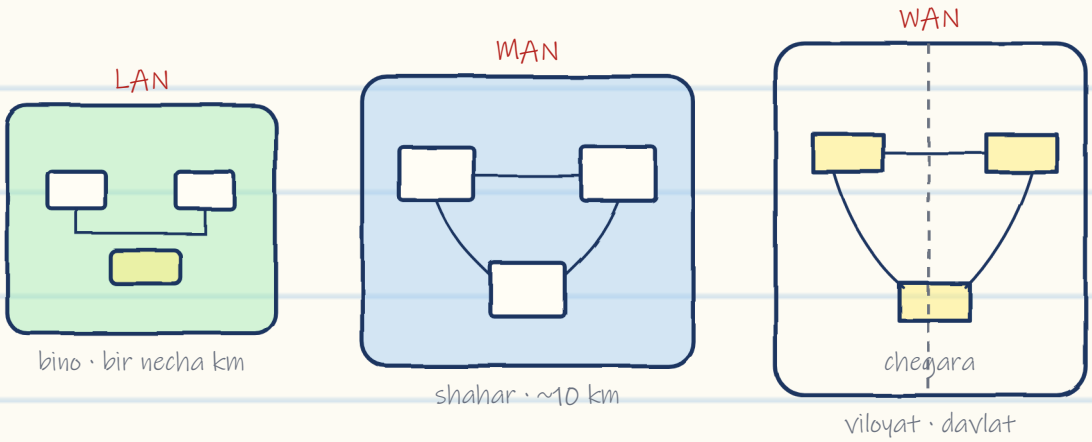
- Resurs almashish — dasturiy (fayl, baza) + apparat (printer, modem)
- Taqsimlangan ishlov — masala bo'linadi, yuk serverlarga taqsimlanadi
- Integratsiyalashgan xizmatlar — data + ovoz + video bitta tarmoqda

3. Rivojlanish bosqichlari



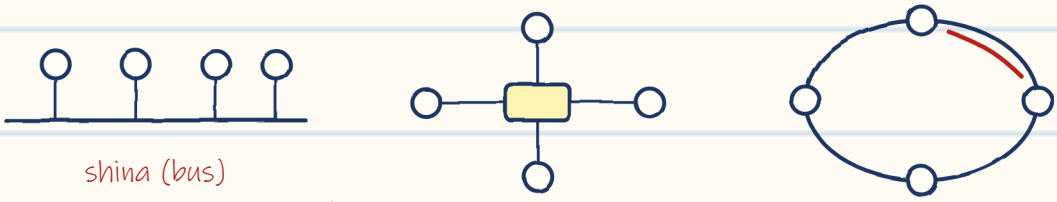
Turtki: kompyuter texnikasi + aloqa texnologiyalari

4. LAN · MAN · WAN



	LAN	MAN	WAN
Qamrov	bino	shahar, ~10 km	davlat
Tezlik	10 Mbit/s – 10 Gbit/s	o'rtacha	pastroq

5. Topologiyalar



shina (bus)

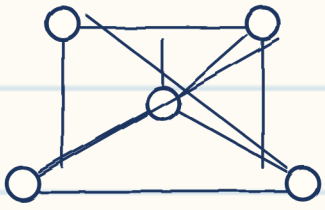
kabel uzilsa — hammasi to'xtaydi

yulduz (star)

halqa (ring)

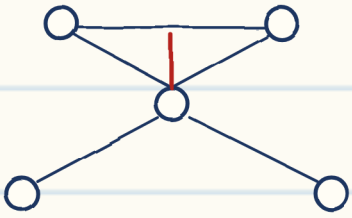
bitta liniya uzilsa — faqat o'sha PC

bir nuqta uzilsa — halqa buziladi



ishonchli, lekin qimmat

to'liq mesh



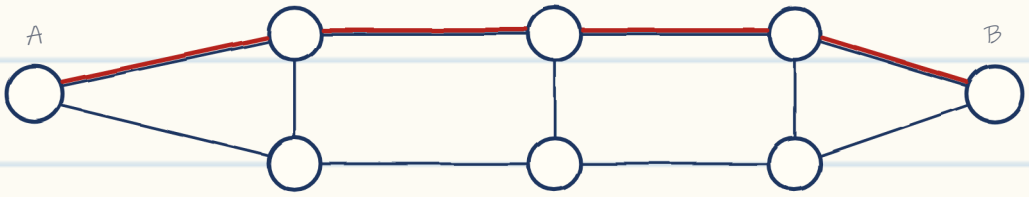
faqat muhim tugunlar — 2 yo'l

qisman mesh

Amalda: LAN — yulduz · magistral va WAN — (qisman) mesh

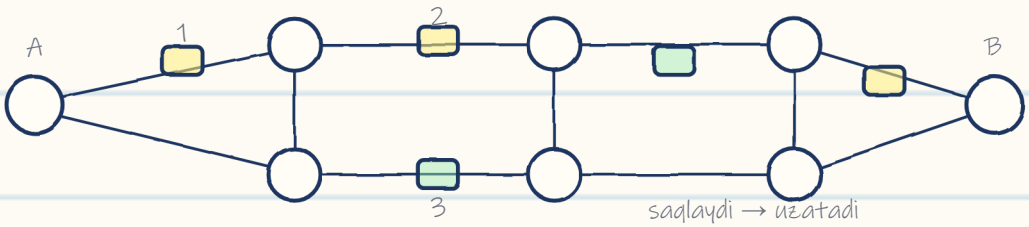
6. Kanal va paket kommutatsiyasi

Kanal kommutatsiyasi — yo'l oldindan band qilinadi



PSTN · suhbat tugaguncha kanal band

Paket kommutatsiyasi — har paket o'z yo'lidan



Kanal

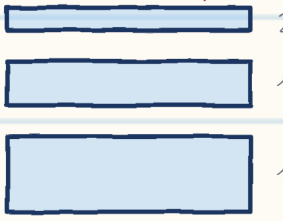
- \+ kechikish kichik, barqaror
- – polosa band turadi

Paket

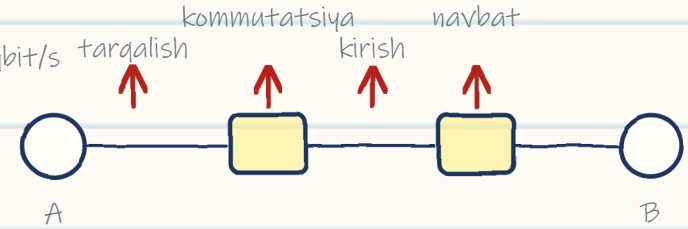
- \+ liniya bo'lishiladi, resurs tejaladi
- – kechikish o'zgaruvchan

7. O'tkazuvchanlik va kechikish

o'tkazuvchanlik = quvur eni



bit/s → kbit/s → Mbit/s → Gbit/s

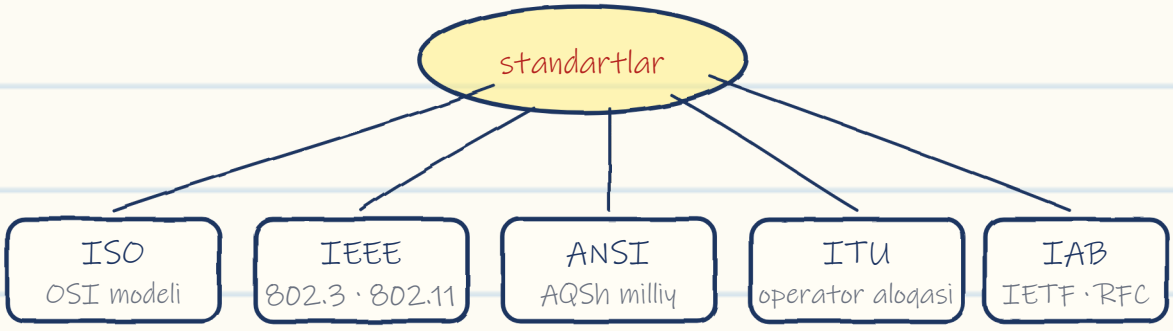


kechikish = shu vaqtlar yig'indisi

SDH	OC-3	OC-12	OC-48	OC-192
Tezlik	155 Mbit/s	622 Mbit/s	2,5 Gbit/s	10 Gbit/s

Kechikishni yo'qotib bo'lmaydi — faqat kamaytiriladi.

8. Standart tashkilotlari



+ EIA/TIA — kabel va ulagichlar (T568A / T568B)

Xulosa

- Tarmoq = bog'langan + mustaqil kompyuterlar
- Vazifa: resurs · yuk taqsimoti · data+ovoz+video
- Yo'l: xost → LAN → WAN → Internet
- Qamrov: LAN (bino) · MAN (~10 km) · WAN (davlat)
- Kanal = band qilingan yo'l, paket = saqlab uzatish
- Ko'rsatkich: o'tkazuvchanlik (bit/s) va kechikish (ms)

Tekshirish

1. Ta'rifdagi 2 kalit so'z? — bog'langan, mustaqil
2. MAN qamrovi? — shahar, ~10 km
3. Ethernet: kim, qachon? — Metkalf, 1973
4. 802.11 nima? — IEEE simsiz LAN